CHAPTER 12

INTERNET AND MOBILE COMPUTING

CHAPTER ESSENCE AT A GLANCE:
Master the global internet infrastructure and wireless mobile cellular network ecosystem powering web services, cloud computing, mobile operating systems, and cybersecurity defenses.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Cellular Frequency Reuse:** Mobile telephony divides geographical areas into hexagonal cells, reusing transmission frequencies across non-adjacent cells to multiply subscriber capacity.

• **Evolution of Mobile Generations:** Cellular technologies advanced from 1G analog voice to 2G digital voice and SMS, 3G mobile data, 4G high-speed all-IP LTE streaming, and 5G ultra-low latency gigabit networks.

• **Internet Service Ecosystem:** The Internet unifies global computing by delivering decentralized services including web browsing, secure transactional ecommerce, email routing, and cloud storage.

• **Proactive Cyber Defense:** Safeguarding digital devices requires defense-in-depth: firewalls, multi-factor authentication, cryptographic protocols (HTTPS and SSL), and vigilance against phishing scams.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Cellular Capacity} \propto \frac{\text{Total Available Frequencies}}{\text{Frequency Reuse Factor } K}$$

Cellular Frequency Reuse Optimization

$$1\text{G (Voice)} \rightarrow 2\text{G (SMS)} \rightarrow 3\text{G (Data)} \rightarrow 4\text{G (LTE)} \rightarrow 5\text{G (IoT and Gbps)}$$

Evolution of Mobile Cellular Generations

$$\text{Mobile OS} \in \{\text{Android (Linux-based, FOSS)}, \text{iOS (Proprietary)}\}$$

Dominant Mobile Operating System Architectures

$$\text{Cyber Defense} = \text{HTTPS} + \text{Firewall} + \text{Antivirus} + \text{2FA}$$

Multi-Layered Information Security Framework

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Differentiate clearly: 1G (analog only), 2G (first digital voice and SMS), 3G (mobile internet), 4G (all-IP LTE packet networks), and 5G (millimeter wave, gigabit throughput, ultra-low latency).

• **Alert:** Do not confuse the Internet (the global physical network of interconnected computers) with the World Wide Web (the service of hyperlinked multimedia documents accessed via HTTP).

• **Alert:** Phishing is fraudulent identity and credential theft via fake websites or emails; Denial of Service (DoS) is intentionally flooding servers with malicious traffic to cause downtime.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between the Internet and the World Wide Web (WWW). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **The Internet:** A global physical infrastructure of interconnected computer networks communicating via the TCP/IP protocol suite. **[1 Mark]**

• **The World Wide Web (WWW):** An information service platform operating on top of the Internet that enables accessing interlinked multimedia documents via hypertext transfer protocol (HTTP). **[1 Mark]**

2. What is the Cellular Concept in mobile communication? Why are cells shaped as hexagons? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Cellular Concept:** Dividing a geographic service coverage area into small physical sectors called cells, each served by its own low-power base station transceiver. **[1 Mark]**

• **Hexagonal Geometry:** Hexagons provide equilateral geometric coverage without gaps or overlaps while approximating a circular radiation pattern with equal adjacent cell boundaries. **[1 Mark]**

3. Trace the key advancements across mobile communication generations: 2G, 3G, 4G, and 5G. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **2G & 3G:** 2G introduced digital voice encryption and SMS text messaging (GSM and CDMA); 3G introduced high-speed mobile data, video calls, and mobile internet browsing. **[1 Mark]**

• **4G (LTE):** Transitioned to an entirely IP-based network architecture providing high-speed mobile broadband, HD video streaming, and low latency. **[1 Mark]**

• **5G:** Delivers multi-gigabit peak data speeds, ultra-reliable low latency communication (URLLC), and massive device connectivity powering the Internet of Things (IoT). **[1 Mark]**

4. Compare the two dominant mobile operating systems: Android and iOS on architecture, licensing, and ecosystem. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Kernel & Licensing:** Android is built on an open-source Linux kernel developed by Google under open-source licensing; iOS is a proprietary, closed-source Unix-based operating system developed exclusively by Apple. **[1.5 Marks]**

• **Hardware Ecosystem:** Android runs on diverse mobile hardware manufactured by various global brands; iOS runs exclusively on Apple iPhone hardware devices. **[1.5 Marks]**

5. Explain major cyber security threats encountered on the Internet: Phishing, Spoofing, and Denial of Service (DoS) attacks. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Phishing:** Fraudulent practice of masquerading as a reputable institution via deceptive emails or fake websites to steal sensitive credentials such as passwords and bank OTPs. **[1 Mark]**

• **Spoofing:** An attacker impersonates another device, email address, or IP address to gain illegitimate trust and bypass network access filters. **[1 Mark]**

• **Denial of Service (DoS):** Maliciously flooding a web server with an overwhelming volume of bogus traffic to exhaust server resources, rendering it unavailable to legitimate users. **[1 Mark]**

6. Explain the working of Global Positioning System (GPS) and list two everyday mobile applications. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Working Principle (Trilateration):** A space-based satellite navigation system; a GPS receiver calculates its precise latitude, longitude, and elevation by measuring signal transit times from at least four orbiting GPS satellites. **[1.5 Marks]**

• **Mobile Applications:** (1) Real-time turn-by-turn road navigation (Google Maps); (2) Location-based services including ride-sharing (Uber) and emergency geolocation tracking. **[1.5 Marks]**

7. State four essential safe computing practices and netiquette guidelines that users should follow when interacting online. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Practice 1 (Strong Passwords & 2FA):** Use complex alphanumeric passwords and enable two-factor authentication on all sensitive online accounts. **[1 Mark]**

• **Practice 2 (Secure Protocols):** Ensure websites use HTTPS with valid padlock certificates before entering confidential credentials or payment data. **[1 Mark]**

• **Practice 3 & 4 (Software Updates & Netiquette):** Keep operating systems and antivirus software regularly updated; maintain polite, ethical communication without forwarding unverified news or cyberbullying. **[1 Mark]**